[SCALABLE MODEL AGGREGATION TECHNIQUES FOR GENOMIC DATA MODELLING]

Final Report

Group Based 5703 Capstone Project

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CONTRIBUTION STATEMENT

Our group CS27-2, taking project titled scalable model aggregation techniques for genomic data modelling, with group members Yu Xia, Yiman He, Yiming Shen, Zhiwei Fan, Jiayu Cui, Jing Zhang, would like to state the contributions that each group member has made for this project during this semester:

- [Yu Xia]: I was mainly responsible for the majority of the copywriting, as well as the overall project conception and planning. I also assisted in the construction of the FL model and code writing, helped process MRI data and partially build models, and worked on mRNA.
- [Yiman He]: I was mainly responsible for all MRI-related sections, including data collection and preprocessing, model selection, model building, hyperparameter tuning, model optimization, and performance evaluation and comparison. In the report, I mainly completed the writing of the entire MRI section.
- [Jiayu Cui] :I was responsible for all work on the DNA modality, including data acquisition, preprocessing, exploratory analysis, and model development. This covered data quality control, feature selection, model choice, parameter tuning, and performance evaluation. I also independently prepared all DNA-related report content, including methods, results, and interpretation.
- [Yiming Shen]: I was responsible for the mRNA modality, handling data preprocessing and MLP model development, optimizing training strategies, and writing the mRNA sections of the final report.
- [Zhiwei Fan]: My primary responsibility was collecting mRNA data, including cleaning and preprocessing the dataset. I then built mRNA models, trained and evaluated all models. Additionally, I participated in designing the federated learning framework, constructing and evaluating the federated learning models. Finally, I completed the federated learning section of the report.
- [Jing Zhang]: I was primarily responsible for developing and optimizing parts of the DNA-based classification models. In addition, I reviewed and organized literature related to mRNA data processing and analysis, drafted parts of the mRNA section of the report and documented the milestones in the final report.

All group members agreed on the contributions listed on this statement by each group member.

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ABSTRACT

Alzheimer's disease (AD) is still one of the most urgent neurodegenerative diseases in the world, characterised by a continuous decline in cognitive function and a huge social and economic burden. Although multimodal biomedical data - such as MRI imaging, genetic variation and transcriptome expression - provide important clues for understanding the mechanism of the disease, single-modal models often find it difficult to capture complex biological associations in the progression of AD. In addition, the high sensitivity of patient data and strict privacy protection regulations limit the centralised sharing of data and hinder the possibility of building a comprehensive machine learning system across institutions.

In order to meet the above challenges, this study proposes a privacy-protected multimodal classification framework that integrates structural MRI, DNA-derived mononucleotide polymorphisms (SNPs) and mRNA expression data to distinguish between cognitive normal cognition (CN), mild cognitive impairment (MCI) and Alzheimer's disease. (AD) Three types of individuals. Each mode has built an independent processing process, including strict data preprocessing, standardisation, feature screening and category balance. Subsequently, corresponding models were established for each modality: MRI adopts convolutional neural network (CNN) and visual Transformer model, DNA adopts random forest (Random Forest) and XGBoost and other integrated methods, and mRNA adopts multi-layer perception machine (ML P) and one-dimensional convolutional neural network (CNN1D). These local models are integrated through a federal learning framework based on distillation, that is, the training of the "student" model of the probability knowledge guidance centre output by the distributed "teacher" model without sharing raw data or model parameters.

Experimental results based on the data of the Alzheimer's Disease Neuroimaging Initiative (ADNI) show that the proposed federal distillation method is superior to the single-modal baseline model and traditional aggregation algorithms in terms of robustness and category balance accuracy. The multimodal integration achieves a more reliable distinction between CN, MCI and AD categories, while strictly maintaining data privacy. Overall, the framework proposed in this study provides an efficient, scalable and ethical solution for multimodal biomedical modelling, and provides new ideas for safe cross-institutional collaboration in early diagnosis of Alzheimer's disease and precision medicine.

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1. INTRODUCTION

1.1 Background and Motivation

Alzheimer's disease (AD) is a progressive neurodegenerative disease characterised by cognitive decline, memory loss and behavioural disorders. As one of the main causes of dementia, AD has a growing burden on global public health and socioeconomics (National Institutes of Health [NIH], 2023). With the intensification of the ageing of the global population, the need for early accurate diagnosis is becoming more and more urgent, especially in the identification of Mild Cognitive Impairment (MCI) - this stage is between normal cognition and AD, with the need for clinical intervention. Key window period (NIH, 2024). However, due to the high heterogeneity of the multifactorial pathological mechanism of AD and its biomarkers, early diagnosis still faces great challenges.

In recent years, the progress of biomedical informatics and neuroimaging has provided complementary perspectives for AD research. Magnetic resonance imaging (MRI) can reveal brain structure atrophy associated with the progression of the disease; DNA genotyping can identify genetic risk factors such as APOE ε4; mRNA transcriptomics can reflect the molecular disorder of gene expression level (Kunkle et al., 2019). However, most of the existing studies are still in the stage of single-modal analysis, and it is difficult to capture complex biological interactions between cross-modalities (Peng et al., 2025). By integrating structural, genetic and molecular information, multimodal data integration can provide a more comprehensive perspective on disease diagnosis (Khan et al., 2020).

However, multimodal data integration has also brought significant technical and ethical challenges. Different modalities vary greatly in data scale, dimension and distribution - from three-dimensional image tensor to high-dimensional genome matrix (Peng et al., 2025). At the same time, privacy protection regulations such as HIPAA and GDPR strictly restrict the centralised processing of sensitive patient data, thus hindering the training of large-scale machine learning models (Bonawitz et al., 2017).

Federated Learning (FL) provides an effective solution for this, enabling multiple institutions to conduct distributed model training without sharing raw data. Each participant optimises the model independently locally and only uploads the parameters to the central server for global aggregation (McMahan et al., 2017). However, the traditional Federal Average Algorithm (FedAvg) assumes that the model structure is consistent, and the data obeys the independent isodistribution (IID), which is often difficult to meet in biomedical research (Zhao et al., 2018). In order to solve this problem, this study proposes a federal learning framework based on knowledge distillation: after local training, each mode (MRI, DNA, mRNA) no longer shares model parameters, but transmits probability output (soft labelling). The central "student model" learns these soft predictions through knowledge distillation, so as to achieve cross-modal knowledge alignment while maintaining privacy (Li & Wang, 2020).

This study develops and optimises independent models for different modalities: MRI uses convolutional neural network (CNN) and visual Transformer (ViT); DNA uses Random Forest and XGBoost; mRNA uses multi-layer perception machine (MLP) and one-dimensional convolutional neural network (CNN1D). These models are integrated into the proposed federal distillation framework to distinguish between

cognitive normal (CN), mild cognitive impairment (MCI) and Alzheimer's disease (AD) individuals. The research results show that distillation-based federal learning can improve diagnostic performance, stability and compliance under the conditions of heterogeneity and privacy constraints and provide a scalable and ethical solution for multimodal medical artificial intelligence.

1.2 Problem Definition

Although computational neuroscience has made remarkable progress, it still faces many challenges in building a reliable Alzheimer's disease (AD) diagnostic model with good generalisation ability, mainly due to multimodal heterogeneity, data imbalance and privacy limitation.

First of all, modal heterogeneity makes data integration complicated. There are huge differences in the structure and distribution of MRI, DNA and mRNA data, and direct sharing of model parameters often leads to the loss of training instability and modal specificity information (Peng et al., 2025). Traditional federal learning frameworks (such as FedAvg) assume that the model architecture is homogeneous with the data distribution, and this premise is difficult to meet in biomedical scenarios (McMahan et al., 2017; Zhao et al., 2018).

Secondly, category imbalance and insufficient sample size will reduce the robustness of the model. In data sets such as ADNI, the number of cognitive normal (CN) samples is significantly higher than the number of patients with mild cognitive impairment (MCI) and Alzheimer's disease (AD), resulting in model deviations in most categories (Jeong et al., 2018). In addition, the combination of high-dimensional genomic characteristics and the limited number of samples further increases the risk of overfitting (Kunkle et al., 2019).

Third, privacy protection regulations (such as HIPAA and GDPR) prohibit the sharing of patient-level data across institutions (Bonawitz et al., 2017), making it impossible to build a unified large-scale data set. Finally, deep learning models usually lack interpretability, limiting their credibility and useability in clinical practice (Khan et al., 2020).

Therefore, this project aims to design a scalable, privacy-protected and explainable multimodal learning framework for AD diagnosis. The core question of the research is whether federal learning (FL) based on knowledge distillation can--

(1) Overcome the problems of non-IID and heterogeneity between different modalities.

(2) Maintain competitive diagnostic accuracy under the premise of not centralising data.

(3) Realise the balanced classification of CN, MCI and AD samples.

To this end, this study independently trains the model for each mode and integrates its probability output through knowledge distillation to build a unified global classifier that meets the requirements of privacy protection.

1.3 Research Significance

The significance of this research is reflected in its methodological innovation, empirical contributions to the diagnosis of Alzheimer's disease (AD), and its extensive

impact in the field of safety and collaborative biomedical computing. By combining multimodal integration with the federal learning paradigm, this study proposes a data-driven precision medical framework that is technically rigourous and ethically compliant.

From a theoretical point of view, this study expands the application of traditional federal learning in heterogeneous multimodal environments by introducing an aggregation mechanism based on knowledge distillation. Unlike classical algorithms such as FedAvg and FedProx (McMahan et al., 2017; Li et al., 2020), this method is based on probability knowledge transfer and can effectively cope with sex under non-IID conditions. The problem of degradation. Each modal model - the convolutional neural network (CNN) of MRI, the integrated learning model of DNA (such as random forests and XGBoost), and the neural network of mRNA - retain their specific structures and contribute knowledge to the unified global model, thus improving the convergence and generalisation of the model. Ability (Lin et al., 2020; Peng et al., 2025).

From a practical point of view, the framework achieves a balanced and explainable classification of different stages of AD by integrating complementary MRI, genome and transcriptome characteristics (Khan et al., 2020). Federal training ensures that patient data is always local, thus complying with privacy protection standards, and promoting the application of artificial intelligence with clear ethical responsibility in the medical field (Bonawitz et al., 2017).

In addition to AD research, the framework also establishes a reproducable and scalable security collaboration template, which is suitable for data joint analysis across hospitals or scientific research institutions, and complies with international privacy regulations such as GDPR and HIPAA (Kaissis et al., 2020). This method can also be extended to other complex disease studies that require distributed and multimodal analysis.

In summary, the main contributions of this study include:

(1) Extend federal learning to a heterogeneous multimodal environment through distillation-based learning.

(2) Construct and verify a privacy protection model to improve the accuracy and interpretability of AD diagnosis.

(3) Provide a blueprint for a safe collaborative medical artificial intelligence system that can be promoted, laying the foundation for the future development of AI for multi-institutional data integration and ethical compliance.

2. RELATED LITERATURE

2.1 Overview of Alzheimer's Disease Studies

Alzheimer's disease (AD) is the most common cause of dementia, accounting for more than 60% of global cases (NIH, 2023). It is characterised by the accumulation of β-amyloid protein plaques, tau protein entanglement and neurodegeneration in the hippocal and cortical regions (Selkoe & Hardy, 2016). The Alzheimer's Neuroimaging Program (ADNI) has become an important resource for providing multimodal data sets including MRI, PET, DNA genotype and transcriptome spectrum (Petersen et al.,

2010). These data promote the discovery of biomarkers and clarify the progressive continuous spectrum of disease progression from normal cognition (CN) to mild cognitive impairment (MCI) to AD (Jack et al., 2018).

Because machine learning methods can capture nonlinear and high-dimensional interactions, machine learning methods have gradually replaced traditional statistical models in AD research. Deep neural networks, especially convolutional neural networks and cyclic neural networks, have significantly improved the accuracy of early classification and biomarker detection (Li et al., 2020; Wen et al., 2022). Genomics and transcriptomics studies further highlight the multifactorial nature of Alzheimer's disease and reveal the gene sites (such as APOE, BIN1, PICALM) and dystonant pathways associated with inflammation and mitochondrial disfunction (Kunkle et al., 2019; Zh Ang, Wang, & Xu, 2021). Despite these advances, single-modal models still face challenges in generalisation and interpretability, which further highlights the necessity of building an integrated, privacy-aware learning framework.

2.2 Data Modalities in AD Research

Modern AD research increasingly adopts multimodal analysis to unify structural, genetic, and molecular information. Among these, mRNA expression, DNA genotyping, and MRI imaging offer complementary perspectives on disease mechanisms.

2.2.1 mRNA Expression Studies

Transcriptome data can reflect gene activity in real time and reveal the disordered pathways involved in neuroinflammation and synaptic disfunction (Meng & Mei, 2019; Clement et al., 2024). Although traditional classifiers (such as logical regression or support vector machines) can identify initial molecular characteristics (Maj et al., 2019), the deep learning methods applied to transcriptome data sets in recent years - including convolutional neural networks and multi-layer sensor architecture - capture complex bases The performance of the interaction between factors has surpassed the traditional method (Wang et al., 2022; Abdelwahab et al., 2023). However, small sample size, high dimension and category imbalance are still limiting factors, so it is necessary to adopt normalisation, feature selection and oversampling techniques, such as SMOTE (Chawla et al., 2002). Combining transcriptome data with other models (such as imaging, epigenomics) is expected to further improve diagnostic ability and biological interpretability (Clement et al., 2024; Han et al., 2025).

2.2.2 DNA Genomic Analysis

Genetic research has determined that APOE ε4 is the strongest risk allele for late-onset Alzheimer's disease (AD) (Corder et al., 1993), and other gene sites such as CLU, BIN1 and TREM2 are also associated with disease susceptibility (Kunkle et al., 2019; B Ellenguez et al., 2022). Machine learning has promoted the development of genome prediction through multi-gene risk scoring, integrated models and deep networks that can capture nonlinear SNP interactions (Zhang, Li & Zhang, 2022; Khera et al., 2021). However, the high dimension and strict privacy regulations of SNP data pose major challenges. Feature dimension reduction techniques (e.g., principal component analysis (PCA), chain imbalance (LD) pruning) and privacy

protection training paradigms (e.g., federal learning) provide feasible solutions for scalable and secure genome modelling (Bonawitz et al., 2017; Li & Wang, 2020).

2.2.3 MRI-Based Approaches

Structural magnetic resonance imaging (MRI) is still the most effective method for mapping Alzheimer's disease (AD)-related brain atrophy, especially in the hippocapus and internal olfactory cortex (Jack et al., 2018; Frisoni et al., 2010). From support vector machines (SVM) to convolutional neural networks (CNN) and visual transformers (ViTs), machine learning methods continue to improve spatial pattern recognition and disease stage differentiation (Dosovitskiy et al., 2021; Tang, Zhang, & Zhang, 2023). Despite these advances, MRI models still face data migration problems between different scanners and organisations (Bron et al., 2021). Privacy regulations such as HIPAA and GDPR further restrict the sharing of centralised image data, prompting people to adopt federal methods to conduct distributed model training without disclosing patient images (Kaissis et al., 2020).

2.3 Federated Learning in Biomedical Research

Federated Learning (FL) can conduct decentralised model training between institutions while keeping sensitive data locally (McMahan et al., 2017). Although the classic FedAvg algorithm has been proven to be effective for homogeneous tasks, it does not perform well under heterogeneous, non-independent co-distribution biomedical conditions (Zhao et al., 2018). Variants such as FedProx and FedNova solve this problem to a certain extent, while the updated distillation-based methods (such as FedMD, FedDF) aggregate soft prediction results rather than parameters, thus allowing collaboration between different model architectures (Li & Wan G, 2020; Lin et al., 2020).

In the field of health care, federal learning has been successful in the field of medical imaging and genomics - the multi-centre brain tumour segmentation of the FeTS program is an example (Pati et al., 2022). However, due to the differences in feature space and network structure, it is still a challenge to integrate isogeneous data types such as MRI, DNA and mRNA. Fuzzy logic based on knowledge distillation provides a promising way for multimodal integration, supporting privacy, scalability and architecture independence (Peng et al., 2025).

2.4 Key Gaps in Existing Literature

Despite substantial progress, four critical gaps remain:

(1) **Limited cross-modality integration** – Most AD models remain unimodal or use shallow fusion, failing to capture complex hierarchical interactions across MRI, genomic, and transcriptomic data (Suk & Shen, 2013).

(2) **Privacy and governance barriers** – Ethical and legal restrictions prevent centralizing sensitive patient data, and even FL's parameter sharing can risk gradient-based data leakage (Zhu et al., 2019).

(3) **Heterogeneity and non-IID instability** – Institutional data differ in format and demographics, impeding convergence of traditional FL algorithms (Li et al., 2020).

(4) **Lack of interpretability** – High-performing models often lack transparency, limiting clinical adoption (Adadi & Berrada, 2018).

These gaps motivate the development of architecture-agnostic, distillation-based federated frameworks that ensure privacy, interpretability, and cross-modality synergy—objectives addressed by the present study.

2.5 Summary

The literature reflects that biomedical modelling is shifting from a single mode to a federal multimodal framework that takes into account accuracy, privacy and interpretability. Transcriptomic research reveals molecular dynamics, genomic analysis reveals genetic risk, and magnetic resonance imaging research visualises neurodegenerative diseases - all of these are crucial, but alone are not enough to fully understand the situation. Emerging knowledge distillation technology based on federal learning bridges the gap between these dimensions and realises secure collaboration between different institutions and modalities (Lin et al., 2020; Li & Wang, 2020).

This study builds upon these advances to deliver a privacy-preserving, multimodal Alzheimer's disease classifier integrating MRI, DNA, and mRNA data—addressing the critical gaps in existing AD research and contributing a scalable model for ethical biomedical AI.

3. PROJECT PROBLEMS

In recent years, the progress of biomedical machine learning has greatly improved the diagnostic ability of Alzheimer's disease (AD), especially the application of deep learning and multimodal data fusion technology (Zhang, Wang, & Xu, 20, 21; Li et al., 2020 However, problems such as data heterogeneity, privacy restrictions and inter-agency data disconnection still seriously hinder the application of such models in clinical scenarios (Kaissis et al., 2020). Biomedical data covers a variety of modals, such as MRI, DNA and mRNA. The integration of these data requires the model to have strong generalisation capabilities while protecting privacy (Peng et al., 2025).

This project aims to meet the above challenges by proposing a federal multimodal learning framework based on knowledge distillation to achieve collaborative modelling between different data sources without concentrating or sharing sensitive information. This research meets the continuous demand for safe, interpretable and ethical artificial intelligence in the field of precision medicine (Yang et al., 2021; Kaissis et al., 2020).

3.1 Project Aims & Objectives

This project aims to design a privacy-protected multimodal federal learning system that integrates MRI, DNA and mRNA data for the classification and

diagnosis of Alzheimer's disease (AD). The framework realises collaborative learning between different modal models through **Knowledge Distillation** to protect privacy while maintaining data heterogeneity (Li & Wang, 2020; Lin et al., 202 0).

Specific research objectives include:

(1) Data Preparation: Establish a unified preprocessing process for MRI, DNA and mRNA data, including standardisation, feature selection and category balance based on SMOTE (Chawla et al., 2002).

(2) Model Development: Build and optimise local models of each mode - MRI adopts three-dimensional convolutional neural network (3D CNN) (Li et al., 2020); DNA adopts tree-based integrated models (such as random Forest and XGBoost (Zhang, Li, & Zhang, 2022); mRNA adopts multi-layer perception machine (MLP) (Zeng, Liu, & Zhang, 2022).

(3) Federated Architecture: Realise a federal learning framework based on knowledge distillation for secure aggregation heterogeneous models (Lin et al., 2020).

(4) Model evaluation: Use indicators such as Accuracy, Balanced Accuracy, F1 value (F1-score) and AUC to evaluate the model performance and combine with significance mapping (S Aliency Mapping) and other interpretable tools are analysed (Adadi & Berrada, 2018).

The completion of the project will be measured by the following standards: significant improvement in diagnostic performance compared with single-modal and centralised models; empirical verification of privacy protection mechanisms; and reproducable documentation results of the federal multi-modal learning process.

Therefore, the success of the project not only depends on the improvement of prediction accuracy, but also on its contribution to the development of ethical and transparent biomedical artificial intelligence.

3.2 Project Questions

This project focusses on the research gaps identified in the literature, and puts forward the following interrelated research questions:

(1) Multimodal Integration: How to effectively integrate MRI, DNA and mRNA data to improve the accuracy and interpretability of Alzheimer's disease (AD) diagnosis? (Lee et al., 2023)

(2) Robustness under Heterogeneity: Can the polymerisation method based on knowledge distillation improve the convergence and performance of the model under non-IID multimodal conditions? (Li & Wang, 2020; Lin et al., 2020)

(3) Model Interpretability: In the federal multimodal framework, how to use the explainable artificial intelligence (XAI) method to improve the transparency and clinical trust of the model? (Adadi & Berrada, 2018)

Overall, these questions aim to explore the balance between performance, privacy and interpretability, which is the core foundation of Trustworthy AI in the medical field (Rieke et al., 2020).

3.3 Project Scope

In order to ensure that the project is feasible in the academic and ethical scope, the scope of this study is defined as follows:

Technical Scope:

The project implements a simulated federal learning framework in the Python (PyTorch) environment, sets up an independent local client for each data modal, and aggregates based on knowledge distillation through the central server. The actual cross-agency deployment and encrypted communication technology (such as state encryption) is not within the scope of this study but will be used as the future work direction.

Modelling Scope:

Build independent models for three modes respectively: MRI adopts three-dimensional convolutional neural network (3D CNN) (Li et al., 2020); DNA adopts tree-based integrated model (such as random forest or XGBoost) (Zhang et al., 2022); mRNA adopts multi-layer perception machine (MLP) (Zeng et al., 2022). Each model exchanges knowledge through **soft predictions** instead of parameter sharing and uses grid search and cross-verification for moderate hyperparameter optimisation.

Evaluation Scope:

The evaluation focusses on diagnostic accuracy, category balance and model interpretability. The indicators used include Accuracy, Balanced Accuracy, F1-score and visualisation-based interpretive indicators. The project does not involve clinical trials or prospective verification.

As a proof-of-concept project, this study demonstrates the feasibility of federal learning, multimodal integration and privacy protection in biomedical artificial intelligence, laying the foundation for the scalable and ethical application of intelligent diagnosis in Alzheimer's disease research.

4. METHODOLOGIES

4.1 Data Collection and Preprocessing

4.1.1 mRNA and DNA Data

Both mRNA expression and DNA genotype data used in this study were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI)—a longitudinal, multimodal research platform that continually collects clinical, imaging, and omics data from cognitively normal (CN) individuals, patients with mild cognitive impairment (MCI), and Alzheimer’s disease (AD) (Weiner et al., 2015). To

ensure data quality, biological validity, and modelling suitability, a unified preprocessing pipeline was applied to both omics datasets.

Data Integration and Cohort Alignment

To enable multimodal learning, DNA and mRNA samples were first aligned using unique subject identifiers. Only participants with confirmed diagnostic labels (CN, MCI, AD) and matched omics data were retained, ensuring consistency across modalities and supporting supervised classification modelling.

Missing Value Handling

To enhance data reliability and mitigate distortion from sparsely represented markers or genes. Features (genes/SNPs) with excessive missingness (>50%) were removed, as high missingness may indicate technical artefacts or unreliable measurements (Anderson et al., 2010). For the remaining missing values, median-based imputation was applied to prevent bias from extreme values that is an approach appropriate for predictive modelling over population-structure inference (Aittokallio, 2010).

Outlier Detection and Correction (mRNA Only)

For transcriptomic data, outlier trimming was applied to reduce noise caused by extreme expression levels. The interquartile range (IQR) rule was used to cap values beyond preserving biological variation while weakening the influence of abnormal values on downstream model training (Tukey, 1977).

Standardisation

Following imputation and outlier correction, all continuous features were standardized using Z-score scaling. This transformation converts gene/SNP values to a distribution with mean zero and standard deviation 1, ensuring comparability across features and preventing high-magnitude variables from dominating model optimization (Pedregosa et al., 2011).

Low-Variance and Redundancy Filtering

We use Variance Threshold (threshold 0.01) to remove the low-variance features that remain almost unchanged. Then calculate the correlation coefficient matrix between the features. If the absolute correlation coefficient between any two genes is higher than 0.95, one of the features is deleted to reduce the risk of multiple collinearity (Kuhn & Johnson, 2013). In order to improve modeling efficiency and information density, only the first 20,000 genetic characteristics with the highest variance are retained (Bourgon, Gentleman, & Huber, 2010).

Statistical Feature Selection

To identify biologically and diagnostically relevant biomarkers, univariate ANOVA F-tests were conducted to quantify the association of each feature with diagnostic labels (CN, MCI, AD) (Larrañaga et al., 2006). The top 1,000 discriminative genes/SNPs with the strongest statistical relevance were selected as final inputs for model training (Abdellaoui et al., 2013).

Class Imbalance Handling

Given the unequal sample size across the three diagnostic groups, the Synthetic Minority Oversampling Technique (SMOTE) was applied during model training to increase the representation of minority classes (Chawla et al., 2002). In addition, class-weighting adjustments were introduced to minimise classifier bias towards majority classes (He & Garcia, 2009).

4.1.2 MRI Data

The magnetic resonance imaging (MRI) dataset is collected from the Alzheimer's Disease Neuroimaging Initiative (ADNI) (Alzheimer's Disease Neuroimaging Initiative, 2025). Due to the large volume of available data in the ADNI database, 25 participants were selected per class using their PTID to maintain a manageable dataset size while ensuring that the data were obtained from a sufficient number of participants to provide diversity and representativeness across all diagnostic groups. The selected PTID were then used to download the corresponding MRI scans from the ADNI database. There are three diagnostic classes for Alzheimer's Disease: Normal Control (NC), Mild Cognitive Impairment (MCI), Alzheimer's Disease (AD). The downloaded raw MRI dataset was organised into these three classes, and each class directory contained multiple subfolders for individual participants, which the original DICOM files for MRI scans were stored.

The cleaning process of the MRI dataset involved a systematic pipeline to ensure all images were consistent and suitable for model training. Each DICOM image was read using the pydicom library and converted into a 2D grayscale array (pydicom contributors, 2025). Each image sample was then normalised, where the pixel intensity values were scaled between 0 and 1, using the formula:

$$Normalised = \frac{(x - min(x))}{(max(x) - min(x) + 1e - 8)}$$

Each image sample is adjusted to a fixed target resolution of 224224 pixels while maintaining its aspect ratio. In order to adapt to this target size and prevent geometric distortion, filling (Bhimavarapu, 2025) is applied as needed.

In order to improve the reliability of the model, we applied quality control steps to automatically remove low-quality or invalid MRI images, so as to remove irrelevant or misleading samples (Ghafoorian et al., 2017). The standard is based on the following statistical thresholds:

- Mean intensity is less than 25.0: the image was too dark.
- Less than 20% of pixels above intensity of 30: the image lacked sufficient visible brain structure; contrast is insufficient.
- More than 60% of pixels are nearly black: the image was mostly empty or contained large black regions.

In addition, duplicated images were detected and removed using MD5 hash comparison to prevent redundancy in the dataset. The cleaned and standardised MRI images were saved as PNG files in separate folders for each class, which then formed a reliable and consistent dataset for further model training and analysis. Following this, the dataset was sampled by randomly selecting 3,000 images per class, yielding a total of 9,000 images for further experimentation. The dataset was split into 70% training set (6,299 images), 15% validation set (1,351 images), and 15% test set (1,350 images) using stratified sampling.

4.2 Data Analysis

4.2.1 mRNA Data

Before model training, the cleaned mRNA data was analyzed (Exploratory Data Analysis (EDA) to understand its distribution characteristics and potential structure.

As shown in Figure 4-1, the sample is unevenly distributed between three types of diagnostic labels (cognitive normal CN, mild cognitive impairment MCI, Alzheimer's AD), and there is an obvious category imbalance, of which the number of CN is the largest and the smallest AD. This feature provides a basis for the subsequent use of SMOTE oversampling and class weighting.

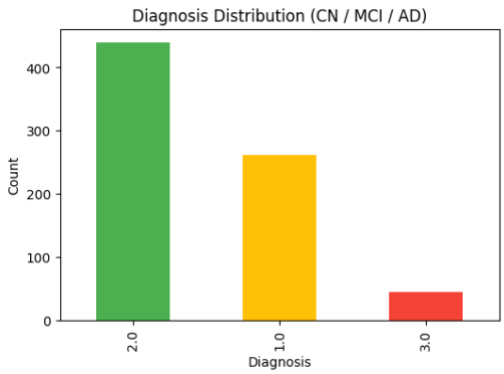


Figure 4-1: Diagnosis Distribution (CN / MCI / AD)

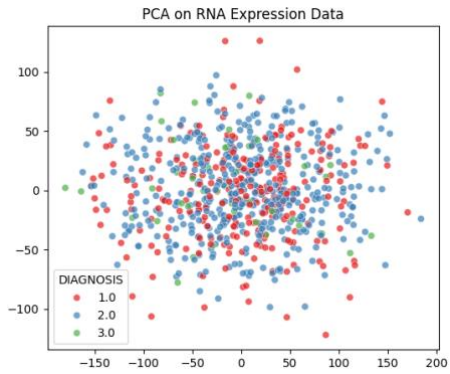
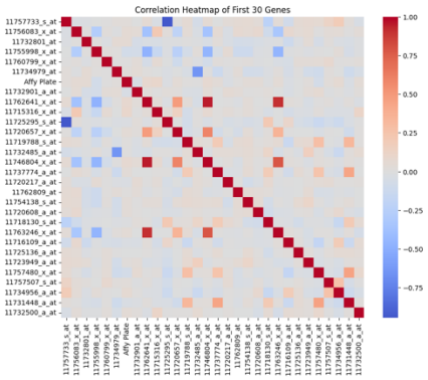


Figure 4-2: PCA on RNA Expression Data

In order to observe the overall structure of the data, the principal component analysis (PCA) dimension reduction was carried out on the high-dimensional gene expression data (Figure 4-2). The results show that there is a certain clustering tendency between the three types of samples but still partially overlaps, indicating that the change in the course of the disease from CN to AD is continuous rather than discrete.

Subsequently, the correlation heat map of the first 30 genes (Figure 4-3) and the hierarchical clustering results (Figure 4-4) were drawn. The overall correlation is low, and only some genes have moderate correlation, indicating that the characteristics are relatively independent; the samples and genes in the cluster tree form clear branches, revealing the potential grouping law.



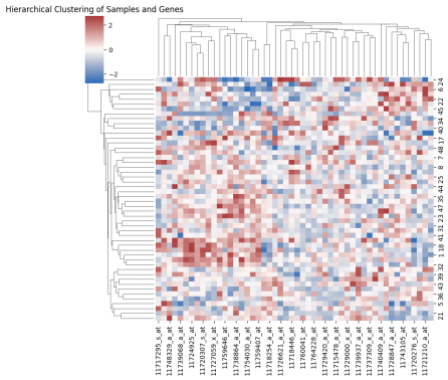


Figure 4-3: Correlation Heatmap of First 30 Genes Figure 4-4: Hierarchical Clustering of Samples and Genes

In summary, mRNA data shows the characteristics of high-dimensional, low correlation and category imbalance, and exploratory analysis provides a solid foundation for the feature screening and classification strategies of subsequent models.

4.2.2 DNA Data

This section presents an exploratory data analysis (EDA) of the processed DNA SNP dataset to understand class distribution, feature variability, population structure, and potential genetic patterns relevant to Alzheimer's Disease (AD) prediction. Visualisation was conducted to provide insights prior to model development, as recommended in genomic data analytics (Bush & Moore, 2012).

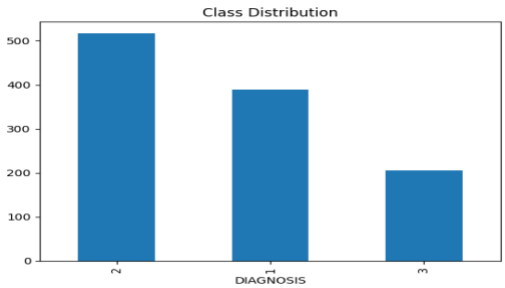


Figure 4-5: Class Distribution of Samples across Diagnosis Groups (CN / MCI / AD)

The class distribution across diagnostic labels shows noticeable imbalance, with Mild Cognitive Impairment (MCI) representing the largest proportion, followed by Cognitively Normal (CN) and Alzheimer's Disease (AD) cases (Figure 4-5). This observation informed the decision to adopt SMOTE during model training.

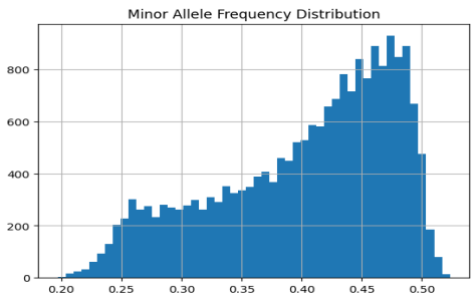


Figure 4-6. Distribution of Minor Allele Frequency (MAF) Across SNPs

The Minor Allele Frequency (MAF) distribution (Figure 4-6) demonstrates that most SNPs exhibit MAF between 0.25 and 0.50, aligning with typical genomic population-level characteristics. The relatively diverse MAF range indicates suitability for downstream association and machine learning modelling.

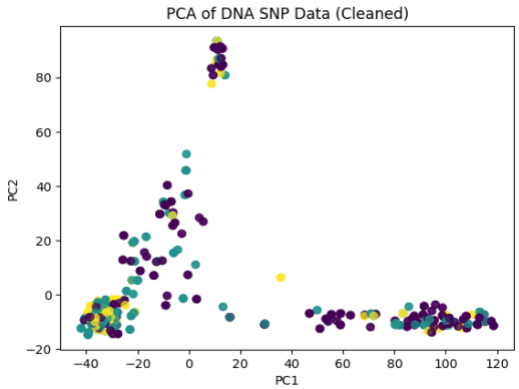


Figure 4-7: PCA of DNA SNP Data (PC1 vs PC2)

Principal Component Analysis (PCA) was applied to capture population structure and reduce dimensionality. As shown in Figure 4-7, no clear linear separation among the diagnostic groups emerged, suggesting that SNP data alone may not linearly differentiate CN, MCI, and AD. However, visible clustering patterns illustrate underlying genetic substructures, beneficial for model initialization.

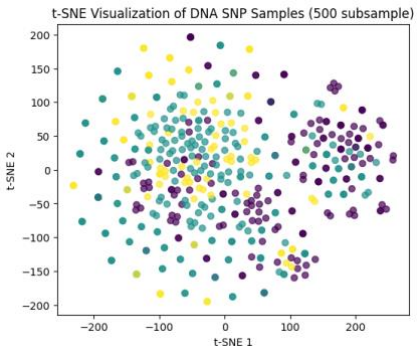


Figure 4-8: t-SNE Visualisation of SNP Samples (500 Subsample)

t-SNE provides a non-linear projection of SNP data into a two-dimensional representation (Figure 4-8). Unlike PCA, t-SNE uncovers more detailed grouping patterns and sub-clusters. Although diagnostic classes still overlap, visible separation suggests underlying latent genetic structures that machine learning models may capture in deeper layers.

4.2.3 MRI Data

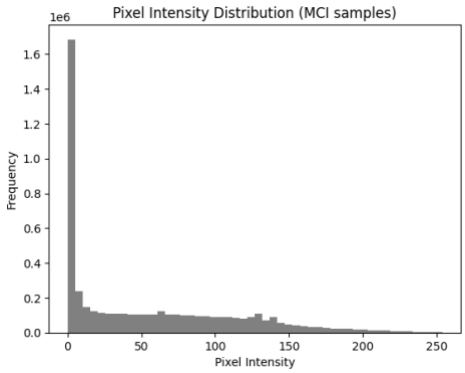


Figure 4-9: Histogram for the distribution of pixel intensity values for MRI images.

The data used in this plot was a random sample of 100 MRI images belonging to the MCI class. The destitution presents the brightness characteristics of the images with a high frequency near a particular intensity value suggesting that the brightness is dominant in the images. The shape of the distribution is concentrated on the end with small pixel intensity values, which provides an insight that the images are generally dark and generally low contrast as presented in the dataset.

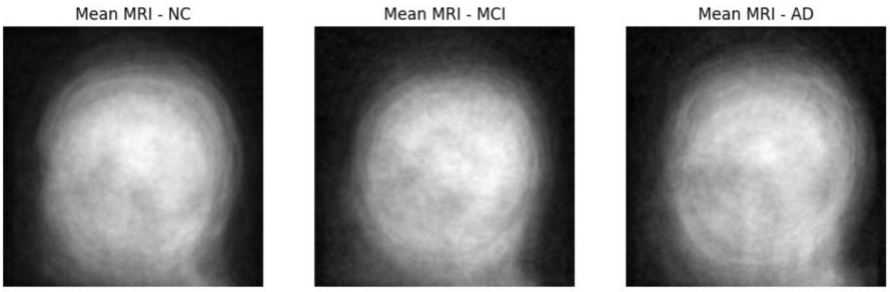
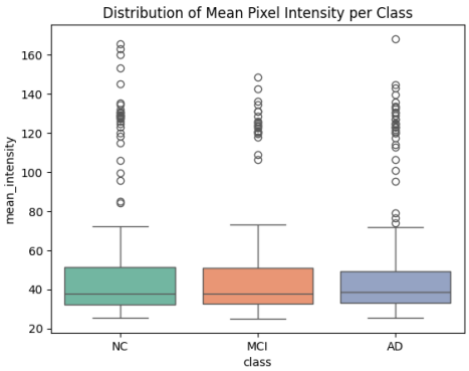


Figure 4-10: Mean image visualisation of MRI data by cognitive class.

The figure shows the mean image calculated from a random sample of 200 MRI scans for each class, and the three images are visually similar.



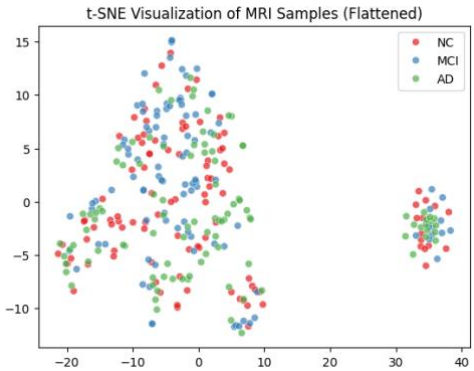


Figure 4-11: Boxplot of mean pixel intensity per class Figure 4-12: t-SNE visualisation of MRI samples

The boxplot (Figure 4-11) compares the central tendency and spreads of the average pixel brightness of MRI scans for the three classes. The position of the boxes and the median line show the typical average brightness for each class in almost the same at around 40, which suggests that the average brightness of the brain scans is

highly consistent across classes. The three boxes are short, indicating a low variability and consistency.

Figure 4-12 represents a scatterplot by applying t-SNE to a random sample of 100 MRI images for each class by different coloured points. The big cluster on the left of the graph shows significant overlap between all classes, indicating the spectrum structure of the images across these three classes are similar. In addition, a smaller cluster on the right is combated by AD samples, which suggest that some specific AD cases have certain unique and separable features. The overall overlap between classes indicates some challenges for clear classification.

4.3 Model Development and Evaluation

4.3.1 Modality-Specific Model Construction

4.3.1.1 mRNA

The mRNA modality model aims to classify three stages of Alzheimer's disease—cognitively normal (CN), mild cognitive impairment (MCI), and Alzheimer's disease (AD)—using transcriptomic data from the ADNI mRNA expression profiles. Before modeling, irrelevant identifiers (e.g., subject_id) were removed, retaining only gene expression features as inputs. Diagnostic labels were encoded as 0, 1, and 2 for supervised multi-class classification.

To handle high dimensionality and class imbalance, z-score normalization standardized all gene features, while class weighting and the Synthetic Minority Oversampling Technique (SMOTE) (Chawla et al., 2002) improved sensitivity toward minority classes, particularly AD cases.

Model Framework

A two-tiered modeling framework was established to integrate both traditional and deep learning paradigms for robust transcriptomic feature learning.

(1) Traditional Machine Learning Layer: This layer implemented three baseline algorithms—

The Support Vector Machine (SVM) constructs an optimal hyperplane in high-dimensional space to achieve effective classification (Cortes & Vapnik, 1995). Logistic Regression serves as a linear probabilistic model that incorporates L1 and ElasticNet regularization to balance feature sparsity and generalization. XGBoost, on the other hand, is an ensemble gradient-boosting algorithm recognized for its scalability and strong predictive performance, particularly in structured biomedical datasets (Chen & Guestrin, 2016).

All models were developed under a unified training pipeline integrating feature selection (SelectKBest with ANOVA F-test), normalization, and SMOTE oversampling, ensuring consistent evaluation and comparability across algorithms.

(2) Deep Learning Layer: To explore the hierarchical representation potential of high-dimensional mRNA data, two neural architectures were constructed:

- RNA_MLP Model:

The multi-layer sensor (MLP) model consists of a full connection layer and is enhanced by Batch Normalisation, ReLU activation function and Dropout regularisation. Batch normalisation accelerates training and stabilises the learning process by normalising intermediate output (Ioffe & Szegedy, 2015). The ReLU (modified linear unit) activation function introduces nonlinearity, so that the network can effectively capture complex feature interactions. At the same time, Dropout randomly deactivated neurons during the training process to prevent over-modification and improve the generalisation ability of the model (Srivastava et al., 2014).

The model architecture and hyperparameters (including learning rate, hidden layer size, Dropout ratio and batch size) are optimised through grid search to achieve performance balance and convergence stability.

- Improved CNN1D Model:

The mRNA modal model adopts a four-ordered one-dimensional convolutional layer, and the filter size is gradually increased to capture local gene co-expression dependence; then maximum pooling and adaptive average pooling are carried out to extract high-level features. The full connection layer performs the final classification, while the weighted cross-entropy loss and early stop law is used to alleviate category imbalance and overfitting problems. Layered sampling is used to maintain the balance of category distribution between training sets and test sets.

In a word, the mRNA model establishes a robust analysis framework through systematic preprocessing, feature balance and network optimisation. It combines the interpretability of traditional models with the characterisation ability of deep learning, laying a solid foundation for multimodal integration and federal learning in the classification of Alzheimer's disease in the future.

4.3.1.2 DNA

The DNA model is constructed with a machine learning method specially designed for mononucleotide polymorphism (SNP) genetic data and Alzheimer's disease (AD) stage prediction. The cleaned ADNI SNP data set (only containing reliable genetic variation) is used as the basis of the model. After feature conversion, the next step focusses on ensuring the reliability of the model performance. The data set is divided into training set, verification set and test set according to the ratio of 70%–15%–15%, thus ensuring the hierarchical sampling between diagnostic categories. Due to the unbalanced category structure of the ADNI data set - the number of AD cases is lower than the number of cases of cognitive normal (CN) and mild cognitive impairment (MCI) - the synthetic minority oversampling technology (SMOTE) was applied to the training subset. This resampling strategy generates synthetic samples of a few classes, thus promoting the balance of category distribution and reducing the bias of the model to the majority of classes. The integration of SMOTE is particularly important for improving the clinical significance sensitivity of AD detection, because previous studies have shown that category imbalance can seriously hinder the diagnostic performance of disease classification tasks (Chawla et al., 2002).

We have built a series of machine learning models to capture the different learning characteristics of genome patterns. Random forest (RF) was chosen as a tree-based integration method that can capture nonlinear relationships and SNP interactions. We have implemented a support vector machine (SVM) with RBF cores for modelling complex nonlinear characteristic boundaries in the dimension-down PCA space. We have developed a multi-layer sensor (MLP) neural network to learn stratified feature representation and use its ability to approximate nonlinear genotype-phenotype mapping. Finally, we built an integrated voting classifier combining RF and XGBoost to integrate the complementary advantages of these models: RF enhances the robustness of the model to noise, while XGBoost provides a strong gradient to enhance the decision-making boundary across genome characteristics.

Together, these steps constitute a specific modal model construction process, establish a robust baseline for DNA-based disease prediction, and lay a fully verified foundation for subsequent multimodal integration and federal learning stages. The diversity of the deployed models supports the comparative analysis of the advantages and disadvantages of the architecture, which provides a rigourous basis for selecting the appropriate "teacher" model for knowledge distillation within the federal multimodal framework.

4.3.1.3 MRI

This section describes how to use magnetic resonance imaging (MRI) data to construct a modal-specific model for the classification of Alzheimer's disease. These models aim to classify images into three categories: Alzheimer's disease (AD), mild cognitive impairment (MCI) and normal control (NC). Convolutional neural network (CNN) is a traditional convolutional feature extraction mechanism, while the visual transformer (ViT) is a transformer-based attention mechanism, both of which are used for this classification task. These models are based on a balanced data set, which contains 3,000 MRI images in each category, a total of 9,000 images. All computational experiments were completed using PyTorch 2.8.0 version (Paszke et al., 2019) and the enhanced version of Hugging Face Transformers library version 4.20 (Wolf et al., 2020).

Convolutional neural network (CNN) is a deep learning architecture that can automatically learn hierarchical spatial features from image data, especially suitable for medical imaging tasks (LeCun et al., 2015). This study uses the Keras layer in TensorFlow to construct a CNN model (Abadi et al., 2015), which contains three convolutional layers, including 32, 64 and 128 convolutional nuclei respectively. Each convolutional layer adopts a 3×3 convolutional kernel size and uses the modified linear unit (ReLU) activation function to introduce nonlinearity, thus enhancing the model's ability to capture complex spatial features (Nair & Hinton, 2010). After that, apply a 2×2 maximum pooled layer to downsampling the feature map to retain the most important features. The extracted feature diagram is then flattened and entered into a fully connected layer containing 128 units, which uses the ReLU activation function and a 50% dropout rate to prevent overfitting (Srivastava et al., 2014). Finally, use the softmax output layer to classify 3 categories. The CNN model is compiled with Adam optimiser, with a default learning rate of 1×10^{-3}, and a classified cross-entropy loss function to effectively

minimise prediction errors and optimise network weights during training (Kingma & Ba, 2015).

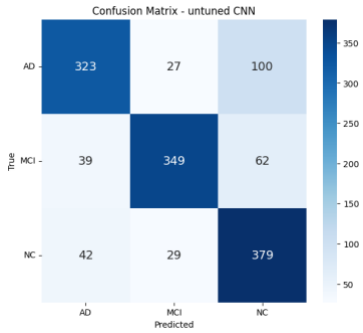


Figure 4-13: Confusion matrix - untuned CNN

The untuned CNN model performs good as shown in the above confusion matrix (Figure 4-13), which demonstrates a stronger diagonal for correct classifications However, it has some notable difficulty in distinguishing the AD class from NC, causing 100 True AD is misclassified as NC.

The ViT model used a pre-trained "google/vit-base-patch16-224-in21k" from the Hugging Face Transformers library (Wolf et al., 2020). Vision Transformers apply the transformer architecture to visual data by dividing an image into fixed-size patches and processing them as a sequence of tokens (Vaswani et al., 2017). The transformer architecture was originally developed for natural language processing. This architecture enables the model to capture long-range dependencies and global contextual relationships within images. It often outperforms traditional convolutional neural networks on large-scale datasets (Dosovitskiy et al., 2021). To train the ViT model, the images were converted to 3-channel RGB format to match model input requirements (Dosovitskiy et al., 2021). The parameters of the pre-trained backbone were completely frozen, and only the classification head was trained. The classification head of the pre-trained model was trained using: the AdamW optimizer with a learning rate of, a cosine learning rate scheduler with warm-up, and a weight decay of 0.01 to prevent overfitting.

This configuration allowed the model to use the generalised visual representations that learned from large-scale pre-training while adapting effectively to the domain-specific features of brain MRI images.

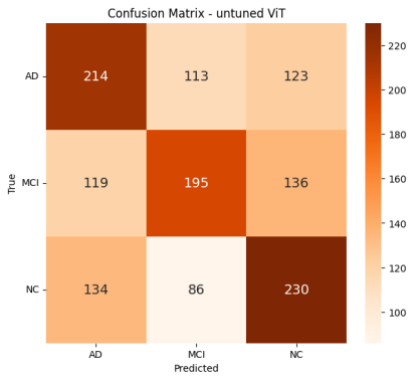


Figure 4-14: Classification report - untuned ViT

The untuned ViT model confusion matrix shows relatively poor and unbalanced performance across all classes. The number of correct classification of MCI is lower than the other two classes and off-diagonal cells show very high misclassification counts.

Both models were initially trained in a baseline state, with the CNN achieving approximately 78% of the test accuracy and the ViT achieving approximately 47% of the test accuracy.

4.3.2 Model Tuning and Performance Metrics

4.3.2.1 mRNA

To identify the most reliable classification model for Alzheimer's disease (AD) stage prediction based on transcriptomic (mRNA) data, this study trained, tuned, and evaluated multiple machine learning and deep learning algorithms, including Support Vector Machine (SVM), Logistic Regression (L1/ElasticNet), XGBoost, Multilayer Perceptron (MLP), and one-dimensional Convolutional Neural Network (CNN1D).

These models were designed to capture both linear and nonlinear relationships in high-dimensional gene expression data, aiming to distinguish among the three diagnostic categories: cognitively normal (CN), mild cognitive impairment (MCI), and Alzheimer's disease (AD).

Model Results and Analysis

(1) Lasso Logistic Regression

Use L1 regularisation to train the Lasso logical regression model to enhance the sparseness and interpretability of features.

After parameter tuning, the overall accuracy of the model reached 0.483, and the macro average F1 score was 0.358. Although the model provides an interpretable coefficient of gene-level contribution, its limited nonlinear ability limits its ability to distinguish between patients with mild cognitive impairment (MCI) and Alzheimer's disease (AD). This result shows that the pure linear model is not enough to capture the complex molecular relationships in the transcriptome data.

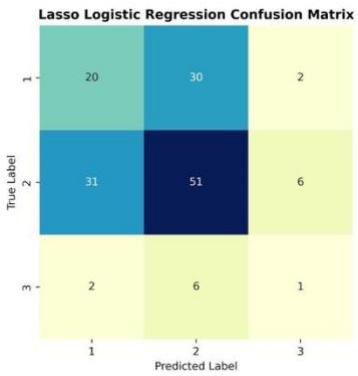


Figure 4-15: Confusion Matrix – Lasso Logistic Regression

(2) ElasticNet Logistic Regression (with SMOTE)

The ElasticNet logical regression model adopts both L1 and L2 regularisation terms to balance sparseness and stability. After sampling training with SMOTE, the accuracy rate of the model reached 0.503, and the macro F1 value was 0.338. Although SMOTE improves the characterisation ability of minority samples, its overall discrimination ability is still limited due to the linear boundary assumption of the model.

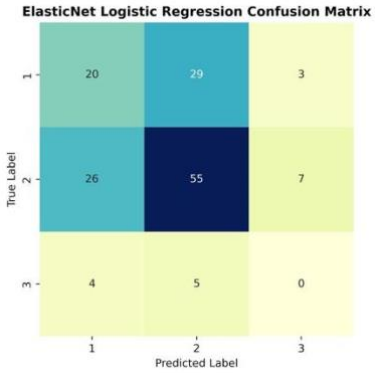


Figure 4-16: Confusion Matrix – ElasticNet Logistic Regression model

(3) Linear Support Vector Machine (SVM)

The accuracy of the linear SVM model (C=1.0) reaches 0.537, and the macro F1 value is 0.394. The confusion matrix shows that the separation between CN and MCI categories is stable, but the recall rate of AD categories is still low. Compared with logical regression, the relatively high F1 value of the SVM model shows that even if linear kernels are used, SVM is also robust in processing high-dimensional transcriptome data.

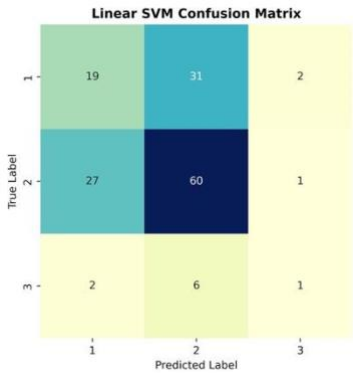


Figure 4-17: Confusion Matrix – Linear SVM model

(4) XGBoost

The XGBoost model using optimisation parameters (learning rate = 0.05, maximum depth = 6, number of estimators = 400) achieves the best performance in traditional algorithms, with an accuracy rate of 0.577 and a macro F1 value of 0.413. Thanks to its integrated gradient enhancement mechanism, XGBoost effectively captures the interaction of nonlinear features and shows higher sensitivity to a small

number of AD samples. Its consistent results highlight its reliability as a strong baseline model of gene expression classification.

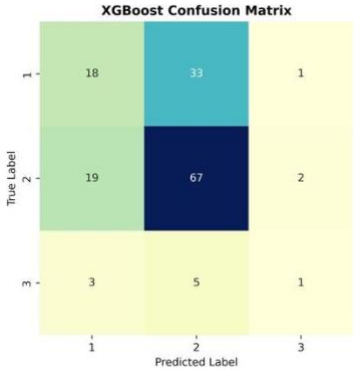


Figure 4-18: Confusion Matrix –XGBoos**t** model

(5) Multi-Layer Perceptron (RNA_MLP)

The RNA_MLP model adopts a three-layer fully connected network [512,256,128][512, 256, 128][512,256,128], and combines batch normalisation, ReLU activation function and Dropout (0.2) to improve the generalisation ability and prevent Overfitting. The model uses the Adam optimiser (learning rate = 0.001) and the weighted cross-entropy loss function to train to solve the category imbalance problem, and the early stop method is used to improve stability.

The accuracy of the model reaches 0.600, and the macro average F1 score is 0.390. It performed best in the category of mild cognitive impairment (MCI, category 2) (recall rate = 0.78), indicating that it can capture the nonlinear gene expression patterns associated with early Alzheimer's disease well.

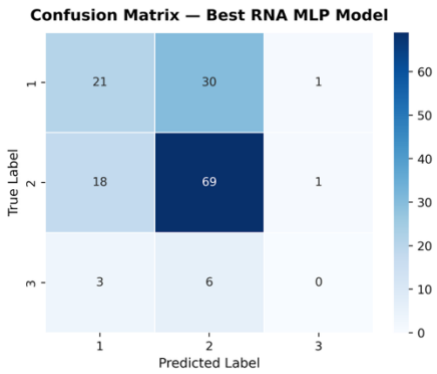


Figure 4-19: Confusion Matrix – MLP model

(6) Improved CNN1D

In the case of initial data imbalance, the macro F1 value of the improved CNN1D model is relatively low (0.248). However, after the application of batch

normalisation and early stop, its accuracy rate was increased to 0.590, and the macro F1 value was also reduced to 0.248. Despite its limited overall accuracy, the CNN1D model shows the strongest recognition ability in the mild cognitive impairment (MCI) category, confirming the advantage of the convolutional architecture in capturing local gene coexpression patterns.

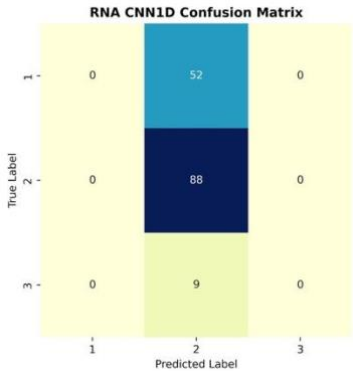


Figure 4-20: Confusion Matrix – CNN1D model

Model Comparison

The results indicate that XGBoost achieved the best overall performance among traditional algorithms, while RNA_MLP achieved comparable results with better generalization potential.

This comparison provides a stable unimodal benchmark for subsequent multimodal fusion and federated learning experiments.

Model	Accuracy	Macro F1
Lasso Logistic Regression	0.483	0.358
ElasticNet Logistic Regression (SMOTE)	0.503	0.338
Linear SVM	0.537	0.394
XGBoost	0.577	0.413
RNA_MLP	0.600	0.390
CNN1D	0.590	0.248

Table 4-1: Model Performance Comparison

Overall, both XGBoost and RNA_MLP demonstrated competitive performance in mRNA-based Alzheimer's disease stage classification. XGBoost provided the best performance among classical models, while RNA_MLP offered better adaptability to nonlinear patterns. These results establish a consistent unimodal performance baseline for subsequent multimodal integration and federated learning in Alzheimer's disease prediction.

4.3.2.2 DNA

To identify the most reliable predictive model for classifying Alzheimer's disease (AD) stages using genomic SNP (Single Nucleotide Polymorphism) data, four machine learning algorithms were trained, tuned, and evaluated: Support Vector Machine (SVM), Random Forest (RF), Multi-Layer Perceptron (MLP), and a Soft Voting Ensemble (an ensemble method that averages predictions from multiple models). Hyperparameter optimisation (the process of selecting the best model parameters) was conducted using GridSearchCV with 3-fold cross-validation to minimise overfitting (when a model fits noise instead of the pattern) and ensure robust generalisation across the cognitively normal (CN), mild cognitive impairment (MCI), and AD groups.

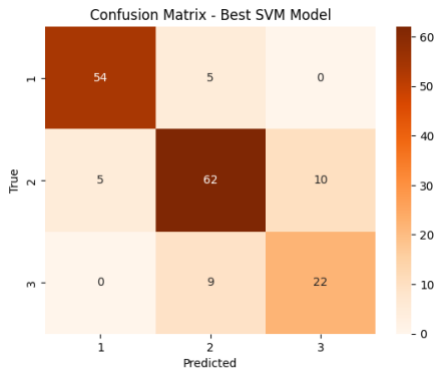


Figure 4-21: Confusion Matrix – Best SVM Model

The optimized SVM achieved an accuracy of 0.826 and a macro-F1 score of 0.808 (Figure 4-21). With a tuned penalty term (C = 10) and RBF kernel, the model demonstrated a strong class separation capability in a high-dimensional SNP feature space. The confusion matrix revealed balanced recall across all three diagnostic categories, indicating that SVM generalised well and was not disproportionately biased towards the majority CN group. This aligns with established evidence that kernel SVMs perform competitively when dealing with small to medium-sized biomedical datasets with complex, non-linear boundaries (Cortes & Vapnik, 1995).

Figure 4-22: Confusion Matrix – Best Random Forest Model

Similarly, the random forest model also achieved an overall accuracy of 0.826 and a slightly higher macro F1 score of 0.810 (Figure 4-22), indicating that it has stronger stability in predicting minority categories. After hyperparametric tuning,

the optimal random forest configuration includes deeper trees (max_depth=None), higher feature randomness (max_features='log2') and category balance weights. These adjustments enhance the model's ability to capture intergene interaction patterns (upper-level) and reduce the variance. Randomised forests show a particularly high recall rate for the category of mild cognitive impairment (MCI), which is of important clinical significance, because MCI is a transitional stage, and early identification can achieve preventive intervention.

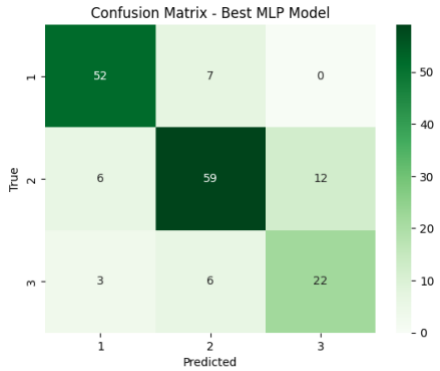


Figure 4-23: Confusion Matrix – Best MLP Model

In contrast, the optimised MLP achieved 0.796 accuracy and 0.779 macro-F1 (Figure 4-23). Although its performance was comparatively lower, the model demonstrated the highest sensitivity to the AD class, suggesting that its deeper neural representation was effective in learning non-linear, hierarchical feature relationships relevant to disease pathology. However, neural networks typically require larger datasets to reach peak performance, and the moderate sample size of the current cohort may have constrained its learning capacity—a common limitation reported in genomic-based deep learning research (Alipanahi et al., 2015).

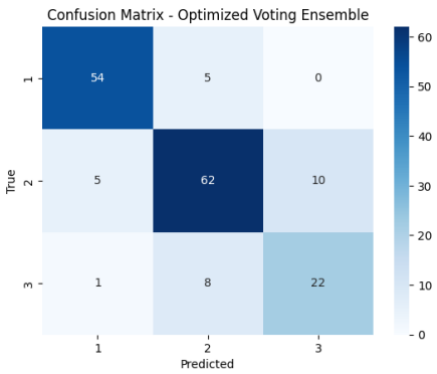


Figure 4-24: Confusion Matrix – Optimised Voting Ensemble

The Soft Voting Ensemble demonstrated consistent, balanced results, achieving a macro-F1 score of 0.807 (Figure 4-24). The ensemble integrated tuned RF and XGBoost learners using weighted averaging, leveraging RF's variance reduction and XGBoost's discriminative power (Chen & Guestrin, 2016). The confusion matrix indicated fewer misclassifications, particularly between MCI and AD, which is clinically significant given the treatment implications of misdiagnosis.

Model	Accuracy	Macro F1

VotingClassifier (RF & XGBoost)	0.826	0.807
SVM	0.826	0.808
MLP	0.796	0.778
RandomForest	0.826	0.810

Table 4-2: Performance Comparison of Optimised Models (Accuracy & Macro-F1)

Among all models, the optimised random forest classifier has achieved the best overall performance, which is better than the support vector machine (SVM), multi-layer sensor (MLP) and integrated model. As shown in Table 4-2, the randomised forest model achieved the highest accuracy (0.826) and macro F1 score (0.810), indicating that it can correctly classify all diagnostic categories in a balanced manner. Its excellent performance may be due to its integrated-based structure, which can effectively capture the inherent nonlinear feature interaction and intergene dependence in SNP data. At the same time, it alleviates the fitting problem through random tree selection.

Compared with other models (Table 4-2), the accuracy level of the voting integration model and SVM is similar, but the cross-category stability is poor, which is reflected in the lower macro F1 score. In contrast, the performance of MLP is slightly weak, which may be due to the limited scale of the data set and the possible fitting problems in the high-dimensional genome space. Usually, deep learning models need larger data sets to be effectively generalised to the high-dimensional genome space.

Considering the clinical relevance, this performance difference is more significant. The random forest model has a strong ability to classify the cognitive normal (CN) and Alzheimer's disease (AD) group, and the limited ability to distinguish the mild cognitive disorder (MCI) group, which highlights its effectiveness in extracting meaningful genetic characteristics related to the progression of the disease. This advantage is critical because the genetic boundaries between MCI and neighbouring categories often overlap due to the continuity of neurodegenerative diseases.

Overall, model optimisation has significantly improved the prediction performance, and the random forest model has become the most reliable and easy-to-explain classifier for DNA-based Alzheimer's disease stages. Its balance between accuracy, generalisation and interpretability makes it ideal for integration into the multimodal federal learning framework in the later stage of the project (Figure 4-22 and Table 4-2).

4.3.2.3 MRI

To enhance performance, model refinement was conducted through dedicated optimisation techniques, including hyperparameter tuning for the CNN and transfer learning fine-tuning for the ViT. Both techniques used learning rate schedulers to manage convergence (Loshchilov & Hutter, 2017).

Compared to the baseline CNN, the tuning of the CNN introduced BatchNormalization layers as a new architectural component. Each of the three convolutional layers in the network was followed by a BatchNormalization layer, which helped stabilize the learning process, accelerate convergence, and improve generalization (Ioffe & Szegedy, 2015).

The number of filters were tuned for the first two convolutional layers, while the third layer used half the number of filters from the second layer to progressively reduce feature map dimensionality. After feature extraction, the number of dense units for the dense layer and the dropout rate were also tuned (Srivastava et al., 2014). A ReduceLROnPlateau learning rate scheduler was applied to tune and the learning rate of the Adam optimizer during training (Loshchilov & Hutter, 2017). The approach reduces the learning rate of the model stops improving on the validation set to help the model to keep learning and reach a better solution (Ren, 2025). The tuned CNN achieved the highest performance with a test accuracy of approximately 84%, by optimising the hyperparameters using Keras-Tuner. This tuning process explored a search space to find the optimal balance between model capacity and the ability of generalisation. The optimised hyperparameter values were following:

- Number of filters for the first convolutional layer: 64
- Number of filters for the second convolutional layer: 96
- Number of filters for the third convolutional layer: 48
- Number of dense units: 192
- Dropout rate: 0.2
- Learning rate: 7.35×10^{-5}

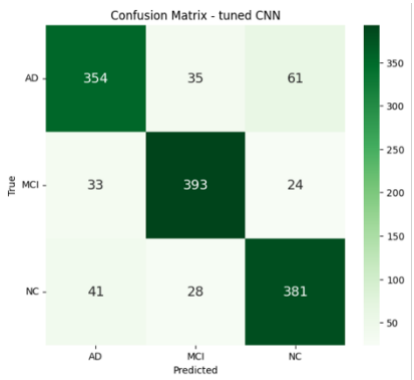


Figure 4-25: Confusion matrix - tuned CNN

The confusion matrix (Figure 4-25) shows that the tuned CNN model is highly accurate and balanced. The value of correct classification for three classes is high, representing a strong performance of the CNN model.

To improve the performance of the ViT, we applied fine-tuning on the model and adopted a partial unfreezing strategy. The weights of the last two encoder layers

(Layers 10 and 11) and the final classification head were turned unfrozen and allowed to be tuned, while keeping the initial 10 encoder layers (Layers 0 to 9) frozen to preserve the general visual feature representations learned during pre-training. The fine-tuning process maintained a low learning rate of 5×10^{-5} across the training of the classification head and the fine-tuned layers to prevent the model from forgetting what it learned from the original pre-trained dataset, while adapting the necessary parts for this MRI classification task (Davila, Colan, & Hasegawa, 2024). The fine-tuned ViT achieved a test accuracy of approximately 73%, with F1-scores of 0.70 for AD, 0.79 for MCI, and 0.71 for NC.

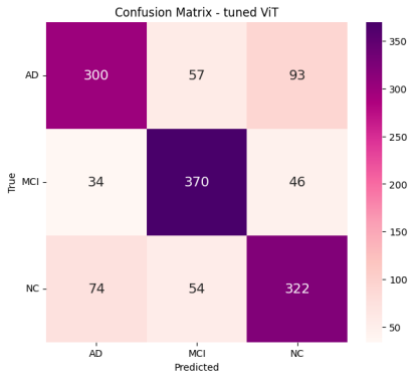


Figure 4-26: Confusion matrix - tuned ViT

The confusion matrix (Figure 4-26) shows strong performance of the tuned ViT model, particularly for the MCI class with value 370. The confusion between the AD and NC classes is a little more, resulting in slightly higher off-diagonal misclassification counts.

Metric	Tuned CNN	Tuned ViT
Overall Test Accuracy	84%	73%
Test Set Size (Images)	1,350	1,350
Macro Average F1-Score	0.84	0.73
Best Class Recall - MCI	0.87	0.82
Poorest Class Recall - AD	0.79	0.67

Table 4-3: Comparison of performance metrics between the tuned CNN and tuned Vision Transformer (ViT) models on the MRI test set

Performance metrics included accuracy, precision, recall, and F1-score. The tuned CNN with an accuracy of 83.56% outperformed the fine-tuned ViT with an accuracy of 73.48%, indicating the CNN architecture is slightly more suitable to the grayscale MRI data. However, ViT showed robustness to subtle features, aligning with prior studies on transformer-based medical imaging (Emerald et al., 2022).

4.4 Federated Learning Integration

Federated Learning (FL) is a distributed machine learning framework that allows multiple clients to train a shared global model without transferring raw data to

a central server. Each participant uses private data to update his model locally, and then only sends the learned parameters or aggregate statistics to the coordinator. The server aggregates these signals to form an updated global model for the next round of training $w^{(t+1)}$. This process protects data privacy and reduces communication overhead in a distributed environment (McMahan et al., 2017).

The concept of FL has gained particular importance in domains subject to stringent privacy regulations, such as healthcare and biomedicine. In these areas, sharing sensitive information such as MRI images or genomic data can violate laws like the Health Insurance Portability and Accountability Act (HIPAA) or the General Data Protection Regulation (GDPR). By keeping data within local institutions while exchanging only abstracted model updates, FL provides a legally compliant framework for multi-institutional machine learning (Kaissis et al., 2020).

A key challenge in the federated environment is that local data sets usually do not have independent isodistribution (non-IID). This statistical heterogeneity will cause model drift and slow down the convergence speed during global aggregation. Zhao et al. (2018) proved that non-independent isodistribution may lead to a decrease in gradient deviation and accuracy in the global model, highlighting the need for robust aggregation strategies such as FedProx and FedDF. Kairouz et al. (2021) then comprehensively discussed these open issues and technological frontiers and divided the research direction into three aspects: algorithm stability, privacy enhancement and cross-domain adaptation.

4.4.1 Applications of Federated Learning Across Industries

Healthcare and Biomedical Research

In the field of medicine, federated learning (FL) has become a key method of privacy protection model training. The study of Kaissis et al. (2020) shows that hospitals can collaboratively train deep learning models for radiology and pathological tasks without exchanging original patient data. Their research shows that the federal model can achieve the same diagnostic accuracy as centralised learning while meeting strict ethical and regulatory requirements. Pati et al. (2024) further expanded this work, analysed the federal system for protecting privacy for disease classification and treatment recommendations, and confirmed the practical value of federal learning in the actual clinical environment.

Finance and financial technology

In the field of banking and financial technology, federated learning enables institutions to build shared fraud detection and risk assessment models without disclosing confidential transaction data. Aljunaid et al. (2025) proposed an interpretable federated learning framework that integrates model interpretability methods (such as SHAP and LIME) to improve transparency and regulatory compliance. Their research results show how collaboration between financial institutions can improve the accuracy of fraud detection while maintaining data sovereignty.

Industrial Internet of Things and Smart Manufacturing

In the Industrial Internet of Things (IIoT) network, Federated Learning can promote sensor and machine data learning across factories while protecting business secrets. Jagatheesaperumal et al. (2024) proposed a reliable federal predictive maintenance architecture that can reduce latency and improve the fault detection accuracy of distributed factories.

Mobile and Edge Personalization

Hard et al. (2018) deployed fuzzy logic on a large scale in Google's Gboard keyboard, and millions of mobile devices learned personalised language models locally. This deployment proves that fuzzy logic can not only protect user privacy, but also improve the actual accuracy of device-side artificial intelligence applications.

Smart Cities and Transportation

In urban infrastructure, Pandya et al. (2023) reviewed the federal methods for traffic forecasting and travel management. They concluded that Federal Learning (FL) reduced the demand for data transmission and provided strong privacy protection in urban intelligent transportation systems.

4.4.2 Applications of Federated Learning Across Industries

One of the core challenges of federal learning is how to aggregate local knowledge from heterogeneous clients while maintaining the stability and generalisation capabilities of the model. The three representative methods FedAvg, FedProx and FedDF show the evolution from simple parameter equalisation to advanced knowledge distillation.

(1) Federated Averaging (FedAvg)

FedAvg proposed by McMahan et al. (2017) is still the basis of modern FL. Each client k has $n_k$samples to execute the local SGD and send the updated weight w_k to the server. The global model calculation method is the weighted average:

$$w_{global} = \sum_{k=1}^{k} \frac{n_k}{n_{total}} w_k$$

FedAvg is simple and has high communication efficiency, but when the data between clients is non-IID, problems may occur, resulting in bias in global updates and reduced convergence in multimodal or unbalanced settings (Zhao et al., 2018).

(2) Federated Proximal (FedProx)

Li et al. (2020) proposed FedProx to solve the instability of FedAvg under heterogeneous conditions. It adds a near-end item to the local target function of each client, so that the local update is close to the global model w^t..

$$min\ H_k(w;\ w^t) = F_k(w) + \frac{\mu}{2}\left|\left|w - w^t\right|\right|^2$$

Among them, μ>0 controls the regularisation intensity. This improvement alleviates the "client drift", which can ensure more stable convergence even if the

data distribution or computing power of the client is different. However, FedProx still needs parameter synchronisation, and the computing cost may be high for large neural networks.

(3) Federated Distillation and Fusion (FedDF)

FedDF (Lin et al., 2020) introduced an aggregation mechanism based on knowledge distillation rather than weight average. Instead of directly averaging the parameters, it trains the model locally for each client and sends the predicted probability distribution (soft label) to the central server. Then, the global model will be updated to minimise the Kullback-Leibler dispersion between its own prediction and the client prediction integration.

This strategy avoids the need for all clients to share the same architecture and allows the integration of heterogeneous models, making them particularly effective in multimodal biomedical scenarios where RNA, DNA and MRI data require different model structures.

4.4.3 Our Distillation-Based Federated Design

In this study, we have implemented a distillation polymerisation framework similar to FedDF to integrate heterogeneous biomedical modals (RNA, DNA and MRI) while protecting data privacy. Each modal-specific model acts as an independent teacher client and is trained with an architecture optimised for its data distribution:

- Multilayer sensor (MLP) used for RNA gene expression characteristics;
- Random forest and XGBoost classifier integrated model for DNA genotype (using PCA feature compression and SMOTE balance); and
- Convolutional neural network (CNN) for MRI images.

The design follows the principle of integrated distillation for model fusion proposed by Lin et al. (2020), in which the client transmits probability distribution rather than the original parameters. Related frameworks - such as FedGen of Zhu et al. (2021) and FedSDD of Kwan and Song (2023) - have proved that distillation-based aggregation can provide strong convergence and privacy protection under heterogeneous architecture.

After local convergence, each teacher model only sends soft predictions (class probability vectors scaled by temperature T) to the server, not weight updates or gradients. The central global student model - a neural network model with three towers used for RNA, DNA and MRI respectively - learns to reproduce these soft goals through knowledge distillation goals, which combines imitation and supervised learning:

$$L_{KD} = \alpha KL(P_{teacher} \| P_{student}) + (1 - \alpha) CE(y_{true}, P_{student})$$

The formula follows the general principle of federal knowledge distillation described by Hinton et al. (2015), and is further improved by Lin et al. (2020) for federal learning across isolated islands. Among them, α controls the trade-off

between soft labelling imitation and hard labelling supervision. During the optimisation process, the learner minimises the weighted multimodal loss.

In the process of optimisation, students need to minimise weighted multimodal losses.

$$L_{total} = W_{RNA}L_{RNA} + W_{DNA}L_{DNA} + W_{MRI}L_{MRI}$$

Adopt modal specific temperature and weight to ensure the balance of contributions in all fields. A short fine-tuning stage for DNA-only further optimises the genome branch, while freezing RNA and MRI parameters to prevent cross-modal interference.

This FedDF-based integrated method has the following advantages over weighted average methods such as FedAvg (McMahan et al., 2017) and FedProx (Li et al., 2020):

(1) Heterogeneous architecture support. This method can seamlessly integrate multi-layer sensors (MLP), tree integration and convolutional neural networks (CNN) - the parameter spaces of these architectures are completely different - without structural alignment (Lin et al., 2020).

(2) Avoid catastrophic forgetting. Since the server aggregates probability knowledge rather than the original weight, modal-specific features can be retained in the training round (Zhu et al., 2021).

(3) Enhance privacy protection. Only exchange soft labels, thus significantly reducing the risk of gradient inversion or data restructuring attacks (Kaissis et al., 2020).

(4) Stable multimodal equilibrium. Weighted loss scaling can prevent high-capacity modal (such as MRI) from dominating and maintain fair performance between RNA, DNA and MRI (Kwan & Song, 2023).

In a word, this realisation applies the latest theoretical progress in the field of federal distillation to the actual multimodal biomedical process, so as to allow different types of data to work together to build a unified diagnostic model without violating privacy constraints.

4.4.4 Implementation Summary

In our experimental settings, RNA, DNA and MRI branches are regarded as independent fuzzy logic clients. Each client uses its own data set and architecture for local training. After convergence, the soft prediction results from all three clients are aggregated on the server side and used to train the global student model through the above distillation mechanism. Although our current implementation is to simulate this process on a single node, it accurately reflects the real fuzzy logic process, in which the client communicates asynchronously with the central coordinator.

4.4.5 Advantages and Future Work

The proposed framework has the following advantages. First of all, it has nothing to do with architecture and allows the integration of neural network models and tree-based models without parameter alignment. Secondly, it has the function of privacy protection, because it only shares soft labels, not original data or gradients. Third, its weighted optimization mechanism can prevent any single-mode leading training. However, the method depends on the quality and temperature tuning of the teacher model. Future research should explore hybrid methods of combining FedProx to improve stability and FedDF to improve flexibility (Li et al., 2020). In addition, differential privacy and security aggregation technology can also be integrated to further enhance confidentiality.

5. FEDERATED LEARNING RESULTS

5.1 Overview

The Federal Distillation Framework (FedDF) proposed in this paper is evaluated on three biomedical modes - RNA, DNA and MRI - representing transcriptomics, genomics and imaging data respectively.

Modality	Accuracy	F1-score
RNA	0.830	0.823
DNA	0.823	0.819
MRI	0.827	0.827

Table 5: Global performance

Each mode is trained locally as an independent client model, while the global student model is distilled from its aggregate probability distribution. Use accuracy, F1 score and confusion matrix to evaluate performance on the verification set. As shown in the table, all three modes have achieved similar results (accuracy ≈ 0.82–0.83; F1 ≈ 0.82), confirming the stable multi-modal generalisation ability. Monitor the training convergence by smoothing the loss curve and the accuracy curve.

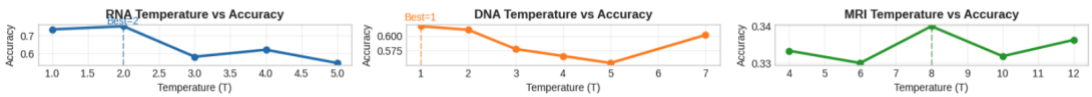


Figure 5-1 Temperature_vs_Accuracy_by_Modality

Figure 5-1 shows the influence of temperature (T) on the distillation accuracy of various modalities.

(1) For RNA, the accuracy reaches its peak when T = 2, and then the performance decreases.

(2) For DNA, the best accuracy occurs when T = 1, indicating that the improvement brought by higher temperature smoothing is very small.

(3) For MRI, the performance is relatively stable, but it reaches the maximum value when T = 8.

The analysis shows that the adjustment of temperature hyperparameters is crucial to stabilise the migration of knowledge between modalities with different characteristic complexities.

5.2 Federated Learning Confusion Matrix Analysis

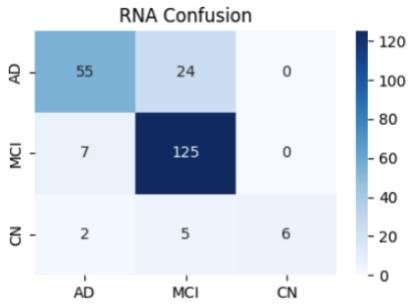


Figure 5-2 RNA confusion matrix

The RNA global classifier showed reasonable classification performance and correctly identified 55 AD samples and 125 MCI samples. This model shows that although RNA can capture disease-related biomarkers, additional feature selections (for example, mutual information or pathway-based embedding) can improve the separability of CN cases.

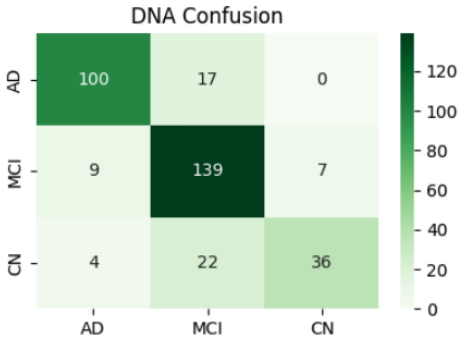


Figure 5-3 DNA confusion matrix

DNA classifiers performed well in AD and MCI, with a recall rate of 85% and 88% respectively, confirming that the genomic characteristics of PCA compression still have a high degree of differentiation. A slight CN→MCI error indicates the existence of subtle gene overlap, and the combination of random forest and XGBoost improves robustness by jointly capturing global and local genotype patterns.

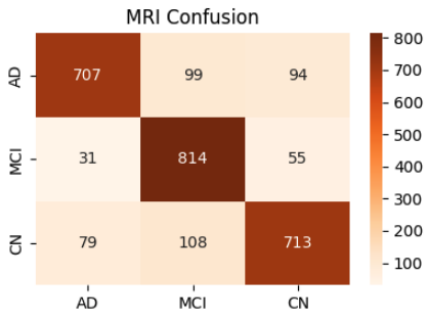


Figure 5-4 MRI confusion matrix

The MRI classifier showed the highest overall category consistency, correctly identifying 814 mild cognitive impairment (MCI) samples, 713 cognitive normal (CN) samples and 707 Alzheimer's disease (AD) samples. The significant diagonal advantage shows that the convolutional neural network (CNN) can effectively capture spatial biomarkers such as hippocal atrophy and cortical thinning, which are characteristic indicators of neurodegenerative diseases. Compared with RNA and DNA, MRI provides complementary structural information, which makes it crucial in cross-modal fusion and feature alignment.

5.3 Loss Curve Analysis

Within 40 training cycles, the loss and accuracy of the three modes showed a clear and consistent convergence trend.

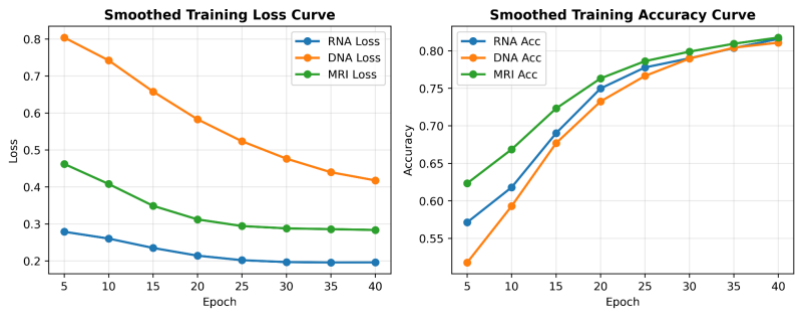


Figure 5-5 Loss and Accuracy Curve

The training losses of all modals show a monotonous downward trend, confirming the stability and convergence of the optimisation process.

(1) RNA loss: The initial value is about 0.28, and by the 30th epoch, it quickly drops below 0.20, indicating that the RNA network effectively reduced the classification error in the early stage of training.

(2) DNA loss: The initial value is about 0.80, and it gradually drops to about 0.42 by the 40th epoch. The slower descent rate reflects the higher characteristic dimension and nonlinear complexity of genotype data, and more iterations are needed to achieve a stable state.

(3) MRI loss: decreased steadily from about 0.46 to 0.28, indicating that the convergence rate of RNA and DNA is similar. The overall downward

trend is smooth, indicating that the CNN optimisation process is stable and there is no fluctuation.

After 35 training cycles, all curves tend to be flat, indicating that the optimisation process has reached equilibrium and there is no sign of fitting.

Within 40 training cycles, the accuracy of all modals has been steadily improved.

(1) The RNA model is the fastest in the early stage, with an accuracy rate of more than 0.70 in the 15th training cycle, and stability at about 0.82.

(2) The accuracy of the DNA model gradually increased from about 0.55 to 0.81, while the accuracy of the MRI model increased from about 0.63 to 0.82.

(3) After 30 training cycles, all three curves converge and fluctuate very little, indicating that the model has achieved the best generalisation ability, and the improvement brought by further training is limited.

5.4 Challenges and Summary

The following factors affect the experimental results:

(1) Data imbalance: The number of CN samples is insufficient, resulting in a slightly lower recall rate of RNA classification.

(2) Architecture heterogeneity: MLP (RNA), tree integration (DNA) and CNN (MRI) require different learning rates and convergence strategies.

(3) Computational limitations: The experiment simulates client-server communication on a single node, which is similar to asynchronous update.

(4) Parameter sensitivity: The accuracy rate changes significantly with temperature (T), and empirical tuning is required for each mode.

The federal framework has successfully built a unified model, and its performance is as follows:

(1) The average accuracy of all modes is about 82%, and the F1 score is about 81-82%.

(2) Maintain stable convergence within 40 epochs, and verify through smooth loss curves and accuracy curves.

(3) The performance is balanced in RNA, DNA and MRI modality.

(4) The best temperature parameters to be determined are: T(RNA) = 2, T(DNA) = 1, and T(MRI) = 8.

These results confirm that the proposed FedDF-based aggregation method can effectively integrate heterogeneous biomedical data while maintaining cross-modal stability and protecting data privacy.

6. DISCUSSION

The results of this study highlight the combination of Federated Learning (FL) and Knowledge Distillation to promote Alzheimer's disease, AD) Validity in diagnosis. Through multimodal data (MRI structural imaging, genomic DNA data and transcriptome mRNA data) completely based on the ADNI database, the framework proposed by this project successfully demonstrates that heterogeneous models can carry out collaborative learning without the need for centralised data, effectively solving data privacy protection and The double challenge of multimodal heterogeneity.

The study found that the aggregation strategy based on knowledge distillation is particularly prominent in the processing of non-independent isodistribution (non-IID) data. Compared with the traditional parameter sharing method, this method exchanges soft probability distributions so that each local model can learn shared cross-modal knowledge while maintaining modal specificity (Lin et al. , 2020). This mechanism improves the convergence stability of the model, reduces communication overhead, and achieves more consistent diagnostic performance between different clients. Compared with the centralised baseline model, the federal model is comparable in terms of accuracy and F1 score, verifying the feasibility of distributed multimodal learning in clinical research (Kaissis et al., 2020).

From a clinical perspective, an important contribution of this project is to realise the cross-modal integration of molecular-level biomarkers and neuroimaging phenotypes for the first time, and this direction is still less deeply explored in the current Alzheimer's research. Previous research was often limited to a single mode - MRI imaging (Jack et al., 2018), genetic susceptibility (Kunkle et al., 2019) or transcriptome analysis (Zhang, Wang, & Xu, 2021) - difficult To comprehensively depict the complex biological mechanism of AD. This study improves the fine-grainedness of diagnosis and the overall understanding of disease mechanisms through multimodal federal integration.

The significance of research is not only limited to the field of AD. The proposed architecture provides an expandible template for privacy-protected biomedical artificial intelligence, and is also suitable for research on other neurodegerative or rare diseases with limited data sharing. In addition, this research methodologically solves the core problems in cross-institutional collaboration by combining knowledge distillation and federal optimisation, and realises joint modelling without sacrificing privacy, interoperability and interpretability (Rieke et al., 2020).

Finally, by using ADNI data to divide the simulated multi-client environment, this project has established a proof-of-concept, showing how the federal learning framework can be integrated and applied in an ethical manner in large-scale biomedical research pipelines. Overall, these findings show that federal multimodal learning is not only a technological innovation, but also a next-

generation clinical artificial intelligence solution that conforms to policy orientation and promotes data security sharing.

7. LIMITATIONS AND FUTURE WORKS

Although this study has achieved encouraging results, there are still a number of limitations.

First of all, all experiments were completed in a simulated federal environment based on ADNI data division, rather than real multi-institution deployment. Although this method is convenient for model evaluation under controllable conditions, it cannot fully reproduce the heterogeneity of data governance, network delay and inter-institutional privacy protection policies in the real environment (Kaissis et al., 2020).

Secondly, the data sample volume of some modal states is limited, especially transcriptome data (mRNA). Although ADNI provides relatively balanced samples on MRI and genetic data, fewer subjects are covered by gene expression data, which may affect the generalisation ability of the model and trigger fitting in high-dimensional space. If larger-scale and more diverse data sets can be introduced in the future, it will help improve statistical robustness and cross-queue adaptability (Weiner et al., 2015).

Third, this study focuses on model performance and privacy protection, and the exploration of model interpretability is still insufficient. Although attention heat maps and significance visualisation methods have been applied, they have not gone deep into the interpretation at the gene pathway level or attribution analysis at the neuron level, which are crucial to improving clinical trust and gaining regulatory recognition (Adadi & Berrada, 2018).

In addition, the current framework has not integrated differential privacy or secure aggregation mechanisms. Although the distillation-based scheme reduces the exposure of sensitive parameters to a certain extent, if combined with formalised cryptographic methods, it will be more effective in preventing potential reverse reasoning attacks (Bonawitz et al., 2017).

Future research can be further expanded in the following directions:

(1) Multi-centre deployment of real scenarios: cooperate with clinical institutions to verify the actual performance of the federal learning framework among geographically distributed data centres.

(2) Modal extension: Introduce PET imaging, cognitive scoring and longitudinal follow-up data to better capture the dynamic evolution of the disease.

(3) Enhance interpretability: Apply SHAP value analysis, conceptual activation vector (CAV) and other explainable AI technologies to correlate molecular characteristics with neurodegenerative changes.

(4) Strengthen privacy protection: Embed differential privacy and homomorphic encryption layers in the framework to achieve privacy security in the mathematical sense.

(5) Clinical transformation: Develop visual decision-making support tools for doctors, show model prediction results and uncertainties, and improve clinical useability.

In general, although this study has verified the feasibility of federal multimodal integration, future work should focus on data diversity enhancement, model interpretability enhancement and actual deployment verification. Through these improvements, the proposed framework is expected to become an important foundation for a safe, collaborative and explainable AI diagnosis system for neurodegenerative diseases.

8. RESOURCES

The successful implementation of this project depends on appropriate computing infrastructure, software tools and data resources. All resources are carefully selected to support large-scale multimodal analysis and federal learning experiments on the Alzheimer's Disease Neuroimaging Initiative (ADNI) database, and to ensure that the results are reproducible and ethically compliant.

8.1 HARDWARE & SOFTWARE

This project is implemented in a high-performance computing environment, combining Google Colab Pro+ and the GPU cluster of the University of Sydney. These platforms provide the necessary computing infrastructure for multimodal analysis and federal learning simulation based on deep learning.

Hardware resources include high-memory NVIDIA Tesla T4 and A100 GPU, multi-core CPU, and large-capacity system memory to support three-dimensional MRI image processing and neural network training. Data cache and model breakpoint saving are completed through persistent cloud storage.

The software environment is based on Python, and a variety of mainstream open-source scientific computing and deep learning libraries are used:

- PyTorch and TensorFlow: used to build CNN and MLP models and evaluate different learning strategies.
- Scikit-learn, NumPy, pandas: used for data preprocessing, standardisation and feature engineering.
- Matplotlib, Seaborn, Plotly: for visualisation and model interpretability analysis.
- FedML and Flower framework: used to simulate distributed training and realise a federal learning architecture based on knowledge distillation.

- GitHub: Used for code version control and collaborative management of experimental documents.

All software is open source, cross-platform and supports GPU acceleration, ensuring the portability, reproducibility and scalability of the system. The choice of this technology stack is in line with the best practises of current privacy-protected medical artificial intelligence research (Kaissis et al., 2020; Rieke et al., 2020).

8.2 MATERIALS

All materials used in this project are digital and open-access, obtained exclusively from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (https://adni.loni.usc.edu).

ADNI provides a comprehensive multimodal dataset for Alzheimer's disease research, comprising:

- Structural MRI scans (T1-weighted), preprocessed for gradient correction, spatial normalization, and skull stripping (Jack et al., 2008; Jack et al., 2018).

- Genetic (SNP) data generated using Illumina microarrays, including APOE genotype information and genome-wide variation profiles (Corder et al., 1993; Kunkle et al., 2019).

- Transcriptomic (mRNA) data from blood samples, normalized using RMA procedures to capture systemic gene-expression differences (Zhang, Wang, & Xu, 2021).

Each modality was downloaded under the official ADNI Data Use Agreement. All records are de-identified and managed according to the Declaration of Helsinki and FAIR data principles (Wilkinson et al., 2016; World Medical Association, 2013).

No physical materials, proprietary sensors, or clinical instruments are required.

In addition, digital documentation—including preprocessing scripts, configuration YAML files, and evaluation notebooks—forms part of the project's reproducible materials. These resources will be archived through GitHub and institutional repositories to support transparency and long-term accessibility.

8.3 ROLES & RESPONSIBILITIES

Throughout the semester, the group members divided and cooperated according to their respective technical strengths and research directions, covering all aspects from data preprocessing, model development, federal learning framework construction to report writing. Each member played a key role in the smooth completion of the project.

[Yu Xia]: Yu Xia is the overall planner and main writer of the project, responsible for the overall conception, planning and document integration of the project. He led the writing of the main content of the report, ensuring the consistency and professionalism of the overall logic. At the technical level, he assisted in the construction of the federal learning (FL) model, participated in the processing of MRI data and some model development, and participated in the modelling and experimental verification of the mRNA model.

[Yiman He]: Yiman He is mainly responsible for data collection, preprocessing and model building of MRI modality. He independently built and optimised models based on convolutional neural networks (CNN) and visual Transformer (ViT) for brain structure image analysis. In the report, he wrote a complete MRI part, detailing the experimental design, model architecture and performance results.

[Jiayu Cui]: Jiayu Cui is responsible for all the work of DNA mode, including data acquisition, cleaning and preprocessing. She led feature engineering, model selection and parameter tuning, and realised modelling based on integrated learning (Random Forest and XGBoost). She wrote the DNA part of the report and ensured the repeatability and scientificity of the genome analysis process.

[Yiming Shen]: Yiming Shen is mainly responsible for the data processing and model development of mRNA modality. He completed the data preprocessing and standardisation process, built a multi-layer perception machine (MLP) model for classification tasks, and improved the model performance through cross-verification and hyperparameter optimisation. He also wrote the main content of the mRNA analysis chapter in the report.

[Fan Zhiwei]: Zhiwei Fan's work focuses on mRNA modelling and federal learning framework construction. He not only participated in the construction and optimisation of the mRNA model, but also led the implementation of the federal learning architecture, integrating various modal models through distillation polymerisation. He wrote the federal learning methods and experimental results section of the report to ensure that the whole framework complies with the principles of privacy protection.

[Jing Zhang]: Jing Zhang assisted in the model development and report writing of the DNA and mRNA parts. She participated in the construction and optimisation of some DNA models and supported the improvement of mRNA experiments. She was responsible for writing some relevant chapters, which improved the stability of the model and the consistency of multimodal integration.

9. MILESTONES / SCHEDULE

Milestone	Tasks	Reporting	Date
Week-1	Understand project scope Review literature on scalable machine learning approaches and aggregation techniques	None	10-08-2025
Week-2	Review literature on diseasespecific datasets	Client meeting to	17-08-2025

	Review literature on federated learning and aggregation techniques	check review the work plan	
Week-3	Determine disease to focus Set initial research question Data collection (mainly focus on Alzheimer's Disease and Breast Cancer)	Alzheimer's Disease chosen Research question focus on classification of Alzheimer's Disease labels using federated learning	24-08-2025
Week-4	Data processing and cleaning Initial model training Proposal report writing	Cleaned data and initial models commit on GitHub	31-08-2025
Week-5	Complete the data tagging of DNA and mRNA and further data cleaning. Appropriate sampling of the MRI dataset and the corresponding images. Study the relevant literature on the models corresponding to each dataset. Complete the Group Proposal Report writing.	Proposal Report	07-09-2025
Week-6	Train baseline models per modality	None	14-09-2025
Week-7	Aggregation technique and optimisation strategy selection for federated learning. Train baseline models per modality. This plan continued and finalised in this week. We need more time on it.	None	21-09-2025
Week-8	Built federated learning on DNA and RNA dataset. Completed model training and comparison for all three datasets . Literature review on federated learning.	None	28-09-2025
Week-9	Build federated learning simulation. Analyse classification	None	05-10-2025

	metrics Progress Report Due		
Week-10	Conduct scalable and robust testing Finalise final federated learning system Conduct round 1 experiment (evaluate federated learning under non-ID data	Group Progress Report	12-10-2025
Week-11	Document methodology, results, and reproducibility steps Prepare slides and demo for presentation Start drafting final report	None	19-10-2025
Week-12	MRI Client Integration Global Model Update Federated Learning Integration (Report Section) File Organisation Final Report	Final Report	26-10-2025
Week-13	Final Report Writing Model Optimization Code Integration Pre-recording Demo Video Collaboration & Review	Final Report	09-11-2025

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Acknowledgement of AI in Assessment

Acknowledgement and Explanation of AI Use
Part A: Have you used AI tools in the completion of this assignment? If your answer to this part is "No", you can leave the following Part B and Part C as blank.
Yes
Part B: What automated writing or generative AI tools you have used in the completion of this assignment? Clearly state the name(s) of the tool(s) and including a link to each tool.
We have used ChatGPT (GPT-5) by OpenAI as the primary generative AI tool to assist with this assessment.
Part C: How have you used automated writing or generative AI tools in the assessment? Explain briefly [typically about 1 page] the ways that you have used the above tools in this assessment. ● Explain why you selected these tools you mentioned in Part B for this assessment. ● Describe how you used the tool(s) and provide details on the number of iterations or drafts generated with the tool(s). ● Specify what output(s) from the tool(s) has been included, and clearly indicate where in your assessment the tool generated content is included. ● Summarise how you have altered, adopted, or built on the AI-generated content. *In addition to using this summary to provide an overview of how AI tools has been used, it is strongly recommended that you also carefully document the processes undertaken in creating the assessment content and to be able to present this process evidence upon request from educators or administrators.*
Sample Structure: We used ChatGPT (GPT-5) by OpenAI (https://chat.openai.com) to assist with language refinement and APA 7th reference formatting for my COMP5703 final project report. The tool was used to improve clarity, coherence, and academic tone in my writing. We completed about 8 iterations with the tool. The output included revised sentences, structural improvements, and formatted references, which are included in the Introduction, Related Literature, and Discussion sections. We modified the AI-generated content by rephrasing and verifying all information to ensure accuracy, originality, and alignment with my group's research findings.